Efficient Facial Image Restoration Using LoRA-Based Finetune

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Abstract

This project explores efficient personalization methods for blind face image restoration using diffusion models. Building on the Gen2Res framework, which finetunes all parameters of a pretrained diffusion model with a small identity specific dataset, Low-Rank Adaptation (LoRA) is investigated as a lightweight alternative. LoRA introduces trainable low-rank matrices into selected layers, enabling personalization with significantly fewer parameters. The impact of dataset size, LoRA scaling factors, and different layer injection strategies on restoration quality are evaluated with PSNR, SSIM, and LPIPS. Experimental results show that LoRA applied to attention layers achieves strong perceptual quality while greatly reducing training time, memory usage, and model size compared to full finetuning. However, extending LoRA to residual or embedding layers yields mixed results and does not consistently improve performance. These findings suggest that attention only LoRA offers an optimal tradeoff between efficiency and restoration quality in personalized diffusion-based restoration.

1. Introduction

Restoring a high-quality image from a degraded version, especially when the type of degradation is unknown, remains a core challenge in computer vision. Unlike traditional restoration methods that rely on specific assumptions about noise or blur, blind image restoration recovers visual content without prior knowledge of how it was distorted. Recently, diffusion models have shown strong performance on such tasks with the capability of modeling the image formation process as a gradual noise injection and learning to reverse it, they can generalize across a wide range of degradation types, making them a compelling solution for blind restoration.

Among related works, Restoration by Generation with Constrained Priors (Gen2Res) offers a novel framework for face image restoration by finetuning a pretrained diffusion model with a personal or generative album of images of the specific character. This constrains the generative space, enabling the model to preserve identity specific features while producing high-quality results without assumptions about the degradation process.

Building on this approach, this project aims to replicate and extend Gen2Res with two goals. First, we reproduce its training and inference pipeline and examine how the number of personal images used for finetune affects restoration performance. This analysis offers insights into the relationship between dataset volume and the generative capability of the finetuned model. Second, we introduce Low-Rank Adaptation (LoRA) as a replacement for full parameter finetuning. By injecting low-rank trainable matrices into the model, LoRA enables efficient personalization with significantly reduced computational cost.

Our results demonstrate that LoRA based finetuning achieves comparable restoration performance while significantly reducing computational and memory overhead. This efficiency enables faster experimentation and easier deployment, making personalized restoration more accessible and scalable.

2. Methodology

2.1 Workflow Overview

This project generally follows the Gen2Res framework, with key modifications to the finetuning process. The pipeline consists of four main stages: preprocessing, training a base diffusion model, finetuning for

personalization, and restoration through sampling.

Before training, facial images are collected and aligned using facial landmarks, then cropped to ensure consistent scale and orientation. These preprocessed images form the dataset for training, finetuning, and restoration.

A base diffusion model is first trained on a large dataset of diverse facial portraits to capture general facial features. Using this base model, personalization is achieved by finetuning with a small set of images from a personal or generative album. While Gen2Res performs full parameter finetuning, this project introduces LoRA, a more efficient alternative that adds trainable low-rank matrices into the model without modifying the base weights.

During restoration, a degraded image is first noised, then progressively denoised using the personalized model. In the Gen2Res setup, only the finetuned model is used. In the LoRA approach, both the base model and the LoRA matrices are applied during sampling. A tunable scaling factor is also introduced to control the influence of personalization during inference.

2.2 Preprocessing

To prepare the image for training, tuning, and restoration, a face alignment process is applied to extract the face from the original image and align it to the center based on facial landmarks. Dlib is used to detect 68 facial landmarks, including eyes and mouth. Using these key points, the orientation and scale of the face are calculated. The face is then rotated to be level, shifted to the center, and scaled to a consistent size. When the face is near the edge of the image, padding is applied by reflecting the image boundaries, allowing the face to be centered without losing any features.



Figure 1. Face Alignment

2.3 Base Diffusion Model Training

The core of the restoration pipeline is built on a Denoising Diffusion Probabilistic Model (DDPM), which learns to reconstruct images by reversing a forward noise process. During training, Gaussian noise is incrementally added to a clean image x_0 over multiple steps t during forward diffusion process, producing a sequence of increasingly degraded image x_t, with the equation:

$$q(x_t|x_0) = \sqrt{\bar{a}_t}x_0 + \sqrt{1-\bar{a}_t}\epsilon \quad (1)$$

Where $\bar{a}_t$ is the cumulative noise schedule, linear or cosine, and ϵ is the random noise added. The goal of training is to learn the reverse process, estimating and removing this noise to recover the original image.

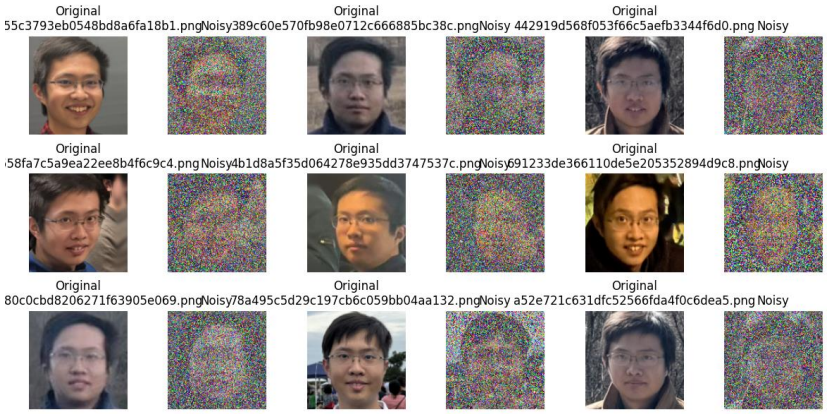


Figure 2. Forward Diffusion Process

The denoising model f_θ is implemented as a UNet based neural network. This architecture is adapted specifically for diffusion models and is conditioned on the time step t, allowing it to process different levels of noise. The model receives a noised image x_t and the current timestep, and predicts the noise $\hat{\epsilon}$ to be removed:

$$\hat{\epsilon} = f_\theta(x_t, t) \quad (2)$$

The UNet model in this project follows an encoder–decoder structure with skip connections, allowing it to preserve high resolution spatial details critical for face reconstruction. Timestep embeddings are injected at various layers to inform the model of the current noise level, enabling it to adjust its denoising behavior accordingly. Attention blocks are also added at multiple resolutions to help the model focus on key facial regions, such as the eyes and mouth, which are essential for identity preservation. This architecture is optimized for denoising tasks and supports multiscale feature processing, making it well suited for high quality face restoration.

To train the model, the goal is to minimize the difference between the predicted noise $\hat{\epsilon}$ and the actual noise ϵ. Two main loss functions are used for supervision.

Mean Squared Error (MSE):

$$L(\theta) = \mathbb{E}_{x_0,\epsilon,t}[\|\epsilon - \hat{\epsilon}\|^2] \quad (3)$$

KL-Divergence Loss:

$$L_{VLB}(\theta) = \sum_{t=1}^{T} D_{KL}(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t)) \quad (5)$$

Model parameters are optimized using the AdamW optimizer with scheduled learning rates:

$$\theta_{t+1} = \theta_t - \eta \cdot \mathrm{AdamW}(\nabla_\theta L(\theta)) \quad (5)$$

The result of this training is a base model capable of reconstructing clean faces from noisy inputs. This model serves as the foundation for later personalization through finetuning.

2.4 Finetuning for personalization

2.4.1 Gen2Res Finetuning

In the Gen2Res research, personalization is achieved by finetuning all parameters of a pretrained base diffusion model using a small set of identity specific images, referred to as a personal or generative album. This process constrains the generative space to preserve identity while still allowing high quality restoration from unknown degradations.

The finetuning follows the same noise prediction objective as base training but uses a much smaller dataset. The pretrained weights are loaded, and the AdamW optimizer is reinitialized with a lower learning rate to prevent overfitting. The model is finetuned for a limited number of steps, typically a few thousand.

To stabilize training, an Exponential Moving Average (EMA) of the model parameters is maintained, and the EMA weights are used during sampling for improved performance. Since all weights are updated, the personalized

model is stored as a separate checkpoint, providing strong identity alignment. However, this approach increases memory usage, training time, and the risk of overfitting on small datasets.

2.4.2 LoRA Finetune

2.4.2.a General LoRA background

Low-Rank Adaptation (LoRA) is a parameter efficient finetuning method that freezes pretrained weights and injects lightweight trainable updates through low-rank decomposition. It builds on the insight that, although large models operate in high dimensional spaces, task specific adaptations often lie in a much lower dimensional subspace. By modeling these updates as low-rank matrices, LoRA efficiently captures the necessary changes without modifying the original model weights.

Instead of updating the full weight matrix $W_0 \in \mathbb{R}^{d*k}$, LoRA adds a trainable component:

$$W = W_0 + \Delta W = W_0 + \frac{a}{r}BA \quad (6)$$

Where $A \in \mathbb{R}^{r*k}$, and $B \in \mathbb{R}^{d*r}$ are trainable LoRA matrices, r is the rank of the decomposition, and a is a scaling factor used to balance update strength and control stability.

The update term $\Delta W = \frac{a}{r}BA$ is added during finetuning and inference. Only A and B are optimized, while W_0 remain frozen.

Thus, for input $x \in \mathbb{R}^k$, the finetuned model computes:

$$Wx = W_0x + \frac{a}{r}B(Ax) \quad (7)$$

This approach decouples personalization from the full model, allowing for modular adaptation with significantly fewer trainable parameters, lower memory usage, and reduced risk of overfitting, which is particularly valuable in low data settings. Unlike adapter layers, LoRA introduces no architectural changes, adds no inference latency when merged, and supports fast task switching by simply swapping the low-rank weights.

To enable efficient and effective personalization, LoRA is selectively applied to key components of the UNet that have the greatest impact on restoration quality. These include attention layers for long range dependencies, residual blocks for local feature processing, and timestep embeddings for noise level conditioning. By focusing on these modules, LoRA finetuning achieves strong identity adaptation with minimal parameter overhead.

2.4.2b Attention Layers: QKV and Output Projection

Attention blocks are central to the UNet's ability to model long range dependencies and preserve semantic consistency in image restoration. Specifically, the query, key, and value (QKV) projections determine how each spatial location interacts with others, making them crucial for retaining facial structure and identity features. In this architecture, each QKV projection is implemented as 1*1 Conv1d layer.

To finetune these layers efficiently, the model replaces standard convolutions with a LoRAConv1d module. This wrapper freezes the original convolutional weights and adds a trainable low-rank update constructed from two additional 1*1 convolutions: lora_down projects the input to a lower-dimensional space $\mathbb{R}^r$, and lora_up maps it back to the output dimension. During the forward pass, the final output is computed as:

$$QKV(x) = Conv_{base}(x) + \frac{a}{r}lora_up(lora_down(x)) \quad (8)$$

This design allows LoRA to inject identity specific information into the attention mechanism while maintaining the efficiency and structure of the original model.

2.4.2c Residual Blocks: Input/Output Convolutions

Residual blocks (ResBlocks) play a crucial role in both feature extraction and reconstruction. They propagate features across the network depth, control local textures and low-level structural details, while preserving spatial resolution through skip connections. Each block includes in_layers and out_layers, which shape information flow across scales and directly impact image fidelity.

To integrate LoRA into the convolutional layers of the ResBlocks, both in_layers and out_layers are wrapped with LoRAConv2d, a module that injects low-rank updates into frozen 1*1 convolutions. Each LoRAConv2d contains a fixed base convolution and two trainable filters: $lora_A \in \mathbb{R}^{r*channel_{in}*1*1}$ and $lora_B \in \mathbb{R}^{channel_{out}*r*1*1}$. During the forward pass, the input is first processed by the base convolution, then passed through a two-step LoRA path:

$$Conv(x) = Conv_{base}(x) + \frac{a}{r} Conv2d(Conv2d(x, lora_A), lora_B) \quad (9)$$

This enables efficient adaptation with minimal parameters, targeting fine grained personalization.

2.4.2d Timestep Embedding Layers

Timestep embeddings are used to condition the network on the current denoising step t, allowing the model to adapt its computation based on the noise level at each stage. These embeddings are essential for guiding the denoising process in a step aware manner, ensuring that the model applies appropriate transformations as it progresses from noisy to clean images. The embeddings are first passed through a multi-layer perceptron (MLP) to increase their representational capacity before being injected into the ResBlocks, where they modulate the feature maps according to the denoising timestep.

To efficiently personalize this time conditioning behavior, LoRA is applied to the linear layers of the MLP in emb layers. The original projection matrix $W_0 \in \mathbb{R}^{d*k}$ is frozen, and a low-rank update is introduced through trainable matrices $A \in \mathbb{R}^{r*k}$ and $B \in \mathbb{R}^{d*r}$. The modified forward pass becomes:

$$MLP(e_t) = W_0 e_t + \frac{a}{r} BAe_t \quad (10)$$

Where e_t is the embedding vector for timestep t.

This implementation uses LoRALinear, a LoRA wrapper around nn.Linear, which efficiently adds the rank constrained update without altering the structure or output dimensions of the original MLP. By adapting the way timestep information is processed, LoRA enables the model to adjust its denoising behavior in an identity aware and data efficient manner.

2.5 Restoration through sampling

2.5.1 Gen2Res Restoration

In Gen2Res, restoration is treated as a generative process guided by the personalized diffusion model. The degraded image y_0 is first projected into the diffusion model's generative space by adding Gaussian noise for a fixed number of steps K, producing a noisy image y_K:

$$y_K = \sqrt{\bar{a}_K} y_0 + \sqrt{1 - \bar{a}_K}\epsilon, \epsilon \sim N(0, I) \quad (11)$$

Through observation, after sufficient noise is added, y_K becomes statistically similar to a clean sample x_K drawn from the model's training distribution. This enables the model to sample x_0, the restored image, by treating y_K as a valid generative input:

$$p(x_0|y_K) \approx p(x_0|x_K) \quad (12)$$

The sampling process begins from y_K and iteratively denoises it step by step until the final output x_0 is produced. This denoising relies entirely on the personalized prior learned during finetuning, without additional

external guidance.

Choosing the right noise level K is essential. If too little noise is added, the input does not align with the training distribution, leading to poor quality samples. If too much noise is added, the identity specific content may be lost. An optimal K balances these effects, ensuring that restored outputs are both high quality and faithful to the original subject.

2.5.2 LoRA Restoration

In LoRA-based restoration, the process begins by adding noise to the degraded image y_0, producing y_K aligned with the base model's distribution. The model then iteratively denoises y_K to reconstruct x_0, using a frozen base model combined with LoRA matrix. During each denoising step, the base model computes the primary feature transformations, while the LoRA adapters introduce small but task specific residual updates. These adapters are selectively inserted into attention layers, convolutional blocks, and timestep embeddings, and are activated during inference without modifying the base parameters. The forward pass at each layer is thus a combination of the original output and the scaled contribution from the LoRA path, controlled by a scale factor a. In practice, this means the model's output at inference is influenced by the personalized identity features encoded in the LoRA weights, while the general structure and generative quality remain anchored to the frozen base. This design allows restoration to benefit from both general diffusion priors and lightweight personalization, enabling identity-consistent restoration with minimal overhead. At inference, only the LoRA scale is adjusted; the base and LoRA weights remain fixed.

3. Experiment

A series of experiments was conducted to compare the original Gen2Res method with various LoRA based configurations. These include evaluations on dataset size, scaling factors, and layer injection strategies. Restoration quality is measured using PSNR, SSIM, and LPIPS, while training time, GPU usage, and model size are used to assess efficiency. Each subsection analyzes a specific factor affecting performance and resource cost.

3.1 Influence of finetuning album size on restoration of Gen2Res

To evaluate how the size of the personal dataset affects restoration in the original Gen2Res finetuning framework, we prepared three training sets containing 5, 15, and 40 aligned face images from the same identity. These were used to finetune all parameters of the diffusion model. For testing, four clean reference images were degraded using Gaussian blur and motion blur, both with a kernel size of 15. The degraded images were then restored using the corresponding fully finetuned models and compared to restored images with base model. Restoration quality was evaluated using PSNR, SSIM, and LPIPS to measure both pixel level accuracy and perceptual similarity.

The result for restoration on Gaussian blurred image:

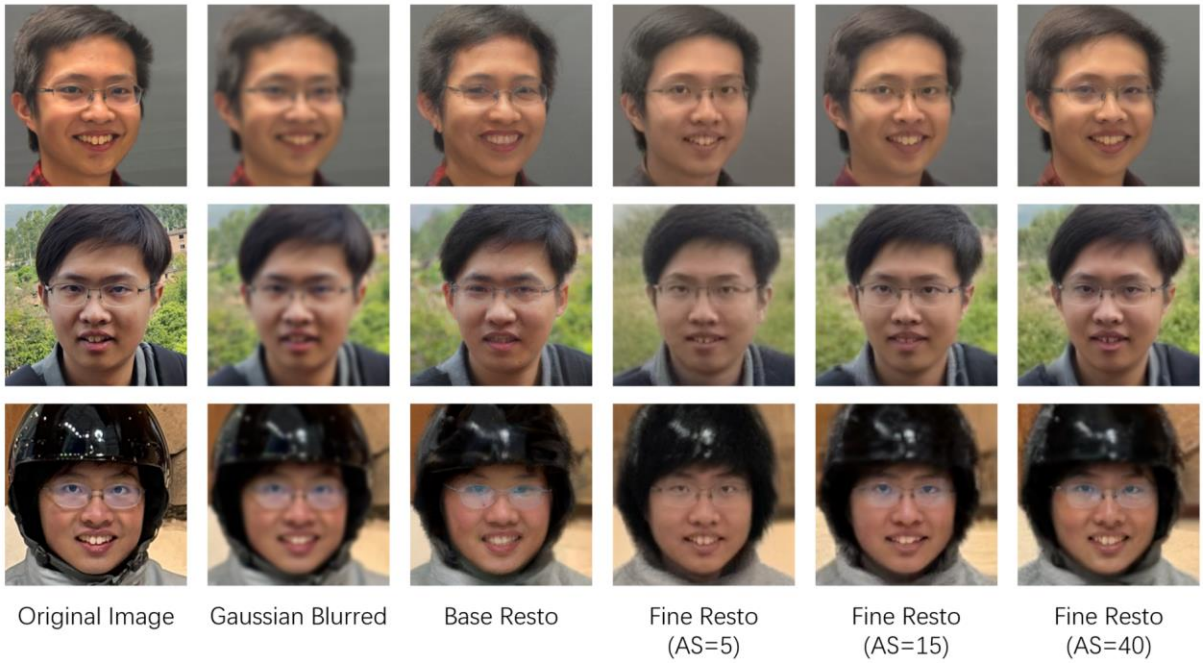


Figure 3. Restoration Comparison on Gaussian Blurred Image

The result for restoration on Motion blurred image:

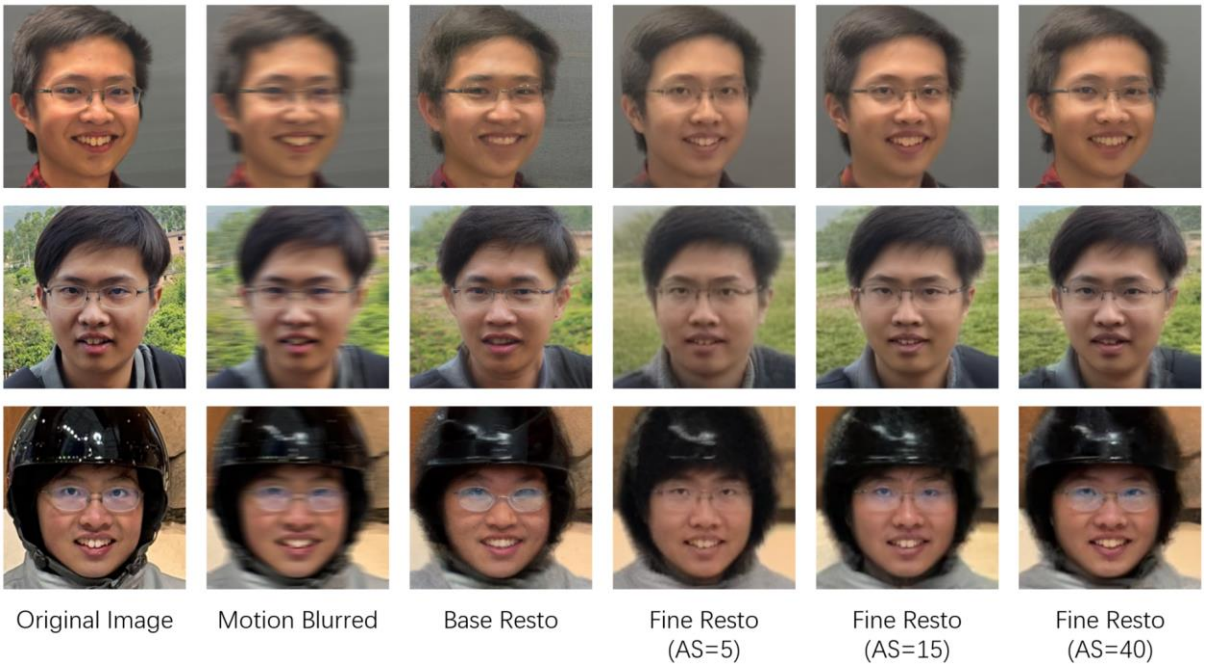


Figure 4. Restoration Comparison on Motion Blurred Image

From the visual results, we observe that as the album size (AS) increases, the restored images progressively resemble the original more closely, with clearer details and improved structure. However, when AS is low (e.g., AS=5), the restored output can appear worse than that of the base model, likely due to underfitting. Beyond a certain threshold, increasing AS yields diminishing returns: AS=40 doesn't offer major improvement over AS=15, suggesting that a moderate number of well aligned images may be sufficient for effective personalization.

The PSNR are calculated for both Gaussian resto and Motion resto:

PSNR For Gaussian Resto

	img1	img2	img3
Gaussian Blurred img	34.23	30.47	31.25
Restored img AS=0	31.82	29.47	29.98
Restored img AS=5	31.08	28.89	29.13
Restored img AS=15	31.95	29.38	29.94
Restored img AS=40	32.51	29.6	30.12

PSNR For Motion Resto

	img1	img2	img3
Motion Blurred img	33.6	30.17	30.91
Restored img AS=0	31.16	29.31	29.82
Restored img AS=5	31.04	28.83	28.91
Restored img AS=15	31.98	29.25	29.76
Restored img AS=40	32.34	29.38	29.99

PSNR generally increases with AS, indicating improved pixel level fidelity. Model finetuned with AS=40 achieves the highest scores. However, AS=5 often underperforms even compared to the base model (AS=0), particularly under motion blur, suggesting underfitting due to insufficient training data. In some cases, even the best restored outputs have lower PSNR than the blurred inputs, highlighting its limitations in capturing perceptual quality.

The SSIM are calculated for both Gaussian resto and Motion resto:

SSIM For Gaussian Resto

	img1	img2	img3
Gaussian Blurred img	0.8677	0.644	0.7053
Restored img AS=0	0.7627	0.5286	0.5611
Restored img AS=5	0.7834	0.5091	0.5264
Restored img AS=15	0.8041	0.5591	0.5909
Restored img AS=40	0.8122	0.5739	0.5991

SSIM For Motion Resto

	img1	img2	img3
Motion Blurred img	0.8454	0.6081	0.6572
Restored img AS=0	0.6991	0.4965	0.5669
Restored img AS=5	0.791	0.4918	0.5157
Restored img AS=15	0.8082	0.5372	0.5692
Restored img AS=40	0.8069	0.5435	0.5935

SSIM shows a similar trend, reflecting better preservation of structural details as AS increase. AS=40 achieves the highest SSIM in most cases, reinforcing the benefit of a larger personalized dataset. Similar to PSNR, AS=5 often yields lower scores than base model, especially for motion blurred images like img2. AS = 15 offers a stable improvement, indicating that sufficient data is needed for reliable structural restoration.

The LPIPS values are calculated for both Gaussian resto and Motion resto:

LPIPS For Gaussian Resto

	img1	img2	img3
Gaussian Blurred img	0.3019	0.4981	0.4236
Restored img AS=0	0.1567	0.3167	0.3264
Restored img AS=5	0.1543	0.4248	0.3551
Restored img AS=15	0.1331	0.328	0.3499
Restored img AS=40	0.103	0.3018	0.3406

LPIPS For Motion Resto

	img1	img2	img3
Motion Blurred img	0.2601	0.4093	0.3624
Restored img AS=0	0.2371	0.3067	0.3211
Restored img AS=5	0.1566	0.445	0.3718
Restored img AS=15	0.1168	0.3094	0.3443
Restored img AS=40	0.1112	0.2873	0.306

LPIPS values decrease as AS increases, indicating enhanced perceptual similarity to the ground truth. The AS=40 model consistently achieves the lowest LPIPS across all test scenarios, aligning closely with human visual perception. AS=15 offers a reasonable tradeoff, while AS=5 tends to degrade perceptual quality.

Overall, increasing the size of the personal album improves restoration quality across all metrics, but only when the dataset is sufficiently large to guide effective adaptation. Very small datasets (e.g., AS=5) often lead to underfitting and, in some cases, worse results than using no personalization base model. In contrast, AS=15 consistently provides a meaningful improvement across PSNR, SSIM, and LPIPS, offering a good balance between performance and data efficiency. While AS=40 delivers the best results overall, AS=15 demonstrates that strong personalization can still be achieved with a moderate number of images. Finally, while PSNR and SSIM measure pixel level and structural similarity, they don't always reflect perceptual quality. Sometimes restored images that look better than the blurred inputs have lower scores. This highlights the value of perceptual metrics like LPIPS for more human aligned evaluation.

3.2 Influence of scaling factor on Attention only LoRA restoration

For attention layer LoRA, from (8) we know the scaling factor a, which would affect the model. In this

experiment, we analyze how the LoRA scaling factor a influences restoration quality when applied only to the attention blocks. All LoRA modules use a fixed rank r=4, while the scale factor a is set to 4, and 8. For training, 100 aligned diverse face images of Hugh Jackman are used to finetune the LoRA adapters. For testing, four blurred and noisy images are generated using gaussian blurred kernel size=25 and noise sigma=0.4. Restoration quality is evaluated using PSNR, SSIM, and LPIPS to assess pixel level fidelity, structural consistency, and perceptual similarity.

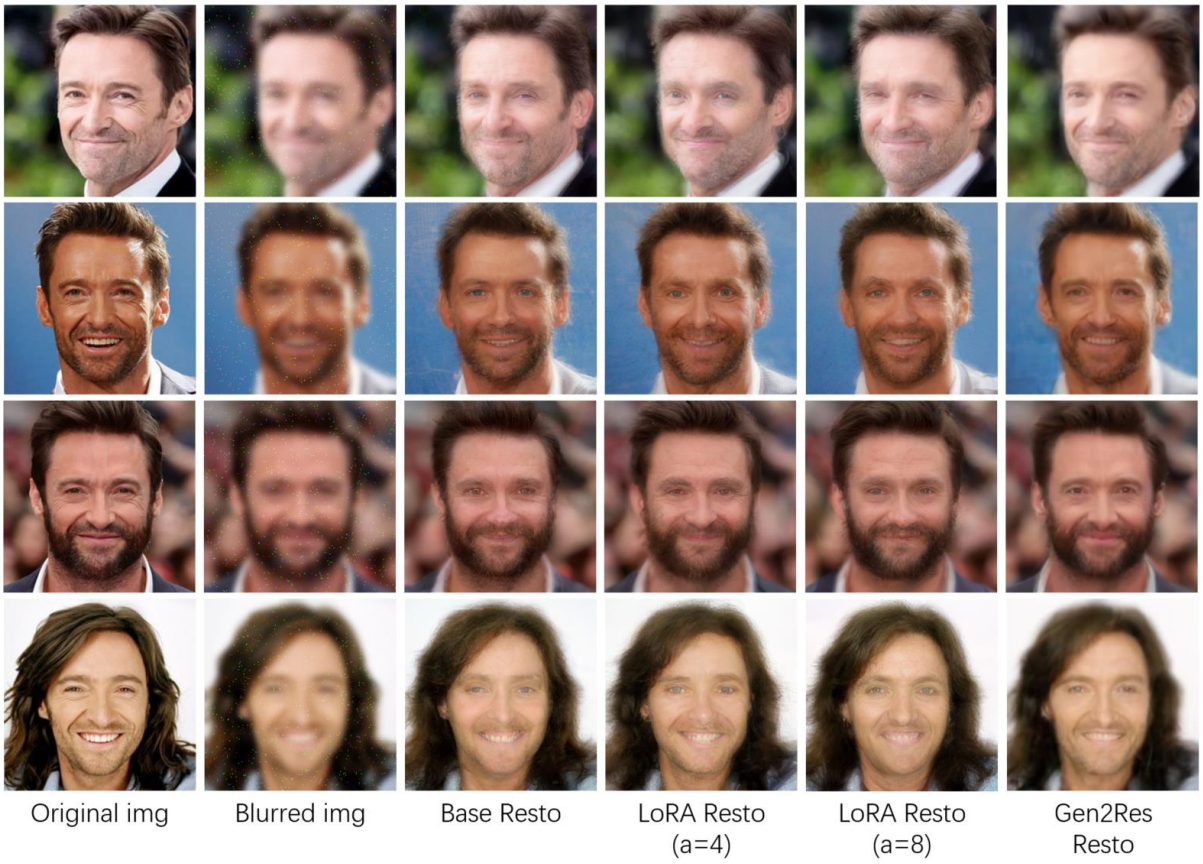


Figure 5. Restoration Comparison for Attention Layer LoRA

From the image comparison above, we observe that restoration with LoRA applied to attention layers recovers more fine details than the base model alone. Features such as wrinkles near the eyes and mouth, as well as the shapes of the eyes, beard, and ears, are more accurately reconstructed. The LoRA enhanced outputs also convey more vivid facial expressions. These improvements generally strengthen as the LoRA scale increases. However, higher scales also lead to smoother textures, which may slightly reduce sharpness. Overall, while LoRA based restoration shows clear improvements over the base model, it still falls slightly short of the full finetuning performance achieved by the original Gen2Res method.

The SSIM, PSNR, and LPIPS values are calculated:

PSNR

	img1	img2	img3	img4
Blurred and noisy img	30.95	31.01	30.77	29.94
Base Resto	29.73	29.92	29.78	29.55
LoRA Resto (a=4)	29.74	30.13	29.68	29.49
LoRA Resto (a=8)	29.66	30.13	29.62	29.54
Gen2Res Resto	29.78	30.35	29.85	29.6

SSIM

	img1	img2	img3	img4
Blurred and noisy img	0.6103	0.5656	0.6097	0.5558
Base Resto	0.6506	0.6015	0.6776	0.5409
LoRA Resto (a=4)	0.6443	0.6175	0.6867	0.5450
LoRA Resto (a=8)	0.6586	0.6295	0.6847	0.5432
Gen2Res Resto	0.6998	0.6890	0.7390	0.5886

LPIPS

	img1	img2	img3	img4
Blurred and noisy img	0.6168	0.5669	0.6364	0.6183
Base Resto	0.2644	0.2889	0.2414	0.3654
LoRA Resto (a=4)	0.2535	0.2574	0.2319	0.3384
LoRA Resto (a=8)	0.2471	0.2409	0.2281	0.3411
Gen2Res Resto	0.2832	0.2663	0.2412	0.3587

The quantitative results show that LoRA attention based finetuning improves restoration performance over the base model, with the effect becoming more noticeable as the scale factor increases.

Across all four images, PSNR for LoRA slightly improve from base model but with small differences LoRA

model slightly improves over the base in some cases but does not surpass the fully finetuned Gen2Res model. Notably, all restored outputs show lower PSNR than the blurred input, highlighting the metric's limitations in perceptual contexts.

SSIM increases consistently with LoRA scaling. Both LoRA settings outperform the base model, with a=8 providing the better scores. Gen2Res still leads across all images.

LPIPS shows the most significant improvement, decreasing as the LoRA scale increases. LoRA with a=8 achieves the lowest LPIPS in most cases, except img4. The socre of LoRA finetune is generally lower than Gen2Res, suggesting LoRA's strength in generating outputs that are visually closer to the ground truth. These results demonstrate that LoRA attention tuning, even with fewer parameters, can enhance structure and perceptual realism, though it may not always reflect in pixel level metrics like PSNR.

It is also worth noting that the current LoRA finetuning setup shows limited generalization. When applied to images from the training set, LoRA with attention layers produces significantly stronger restoration results compared to unseen test images, suggesting that it captures identity specific features well but may overfit even with high volume finetuning set.

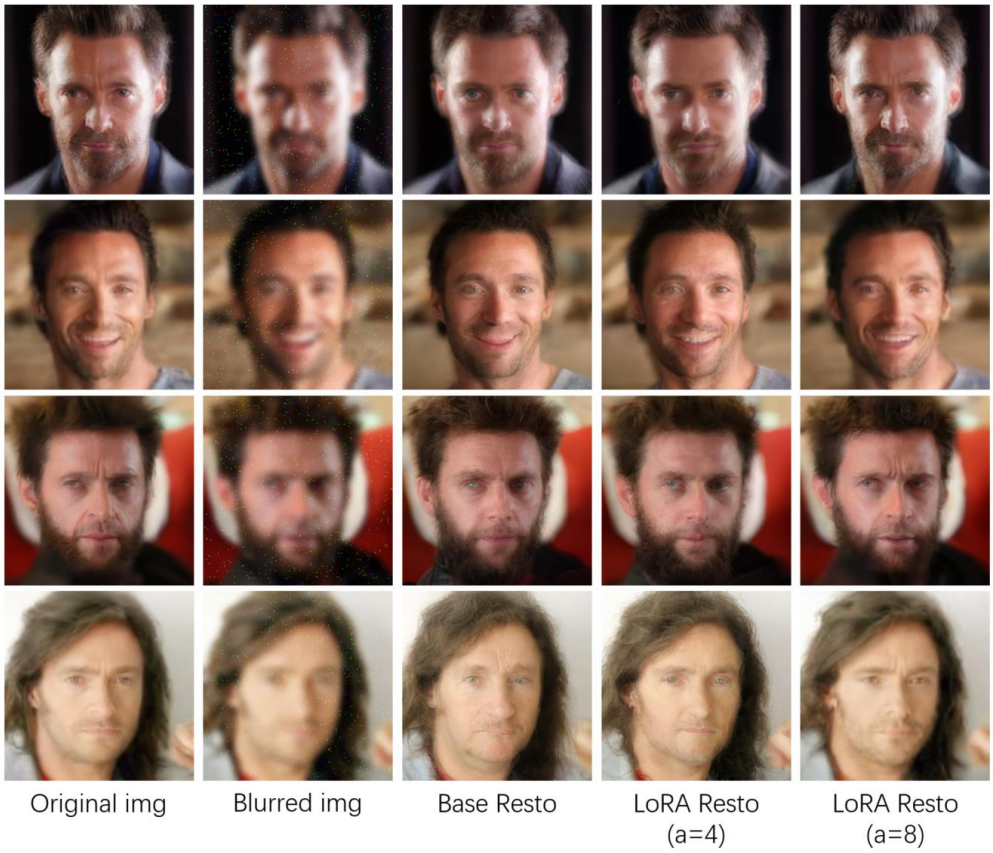


Figure 6. Comparison of restoration on training set

3.3 LoRA restoration with more layer injections

While LoRA applied only to attention layers improves restoration quality, it still falls short of the performance achieved through full finetuning. To investigate whether deeper personalization enhances results, LoRA was further injected into additional parts of the network: the input and output convolution layers of the residual blocks, and the linear layers of the timestep embedding module.

Two new configurations were evaluated, one extending LoRA to the in/out layers, and another to the embedding layers, each building on the original attention only setup. These components were tested separately to

isolate their individual effects and were not used simultaneously.

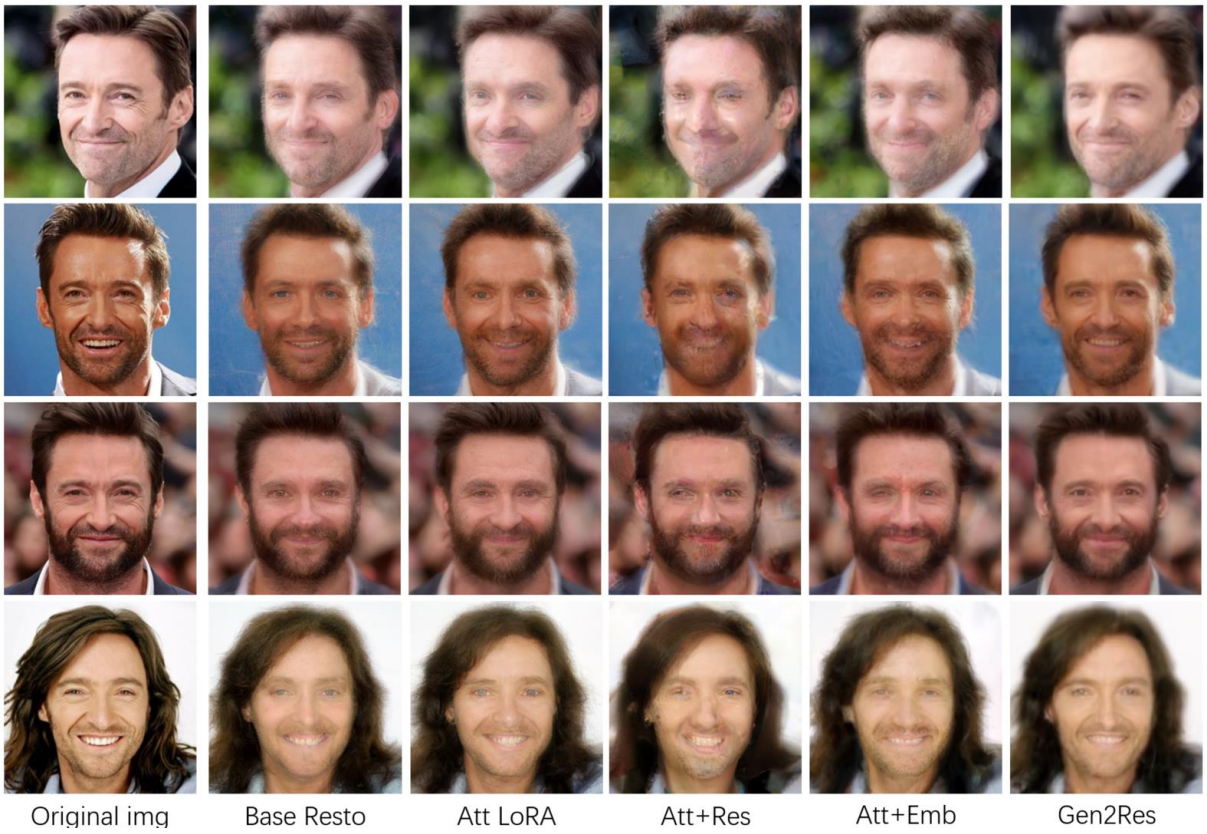


Figure 7. Restoration Comparison for Different LoRA Layer Injection

Visually, the results from injecting LoRA into additional layers were less promising than expected. The configuration with LoRA applied to the residual block's in/out layers exhibited artifacts and slight overlap in facial features, suggesting instability in the reconstruction. Meanwhile, the embedding layer injection resulted in outputs that appeared smoother but lacked sharpness and clarity, indicating weaker restoration of fine details. These outcomes suggest that simply expanding LoRA injection to more components does not guarantee better performance and may introduce new challenges in maintaining identity fidelity and image sharpness.

The SSIM, PSNR, and LPIPS values are calculated:

PSNR

	img1	img2	img3	img4
Blurred and noisy img	30.95	31.01	30.77	29.94
Base Resto	29.73	29.92	29.78	29.55
LoRA Att	29.74	30.13	29.68	29.49
LoRA Att+Res	29.46	30.23	29.59	29.52
LoRA Att+Emb	29.71	30.27	29.7	29.51
Gen2Res Resto	29.78	30.35	29.85	29.6

SSIM

	img1	img2	img3	img4
Blurred and noisy img	0.6103	0.5656	0.6097	0.5558
Base Resto	0.6506	0.6015	0.6776	0.5409
LoRA Att	0.6443	0.6175	0.6867	0.5450
LoRA Att+Res	0.6224	0.6324	0.6569	0.5592
LoRA Att+Emb	0.6522	0.6499	0.6837	0.5594
Gen2Res Resto	0.6998	0.6890	0.7390	0.5886

LPIPS

	img1	img2	img3	img4
Blurred and noisy img	0.6168	0.5669	0.6364	0.6183
Base Resto	0.2644	0.2889	0.2414	0.3654
LoRA Att	0.2535	0.2574	0.2319	0.3384
LoRA Att+Res	0.2852	0.2460	0.2324	0.3036
LoRA Att+Emb	0.2604	0.2750	0.2584	0.3067
Gen2Res Resto	0.2832	0.2663	0.2412	0.3587

Adding LoRA to the residual block in/out layers yields PSNR values similar to the attention only configuration. However, the visual quality is noticeably worse, even in cases where the PSNR is slightly higher, suggesting that pixel level metrics may not fully reflect perceptual fidelity. The embedding layer variant shows marginal improvement over Att+Res and slightly higher PSNR than attention only in some cases, but overall, the results are comparable. Visually, it performs similarly to or slightly worse than attention only in most examples.

SSIM improves noticeably with LoRA Att+Emb, surpassing attention only in most cases. The Att+Res variant shows fluctuating SSIM values, sometimes higher and sometimes lower, but again, these structural scores do not reliably align with the actual visual quality of the restorations.

LPIPS, which reflects perceptual similarity, favors the attention only model, which achieves the lowest scores, even outperforming Gen2Res. Adding LoRA to residual blocks increases LPIPS. LoRA Att+Emb performs

slightly better than the base model in LPIPS but does not exceed the attention only setup, indicating limited perceptual improvement from timestep embedding injection alone.

Among all LoRA configurations, injecting only into the attention layers yields the best overall balance of structural and perceptual quality. Extending LoRA to residual block convolutions introduces visual artifacts, while adding it to embedding layers brings slight SSIM improvements but no clear perceptual gain. Overall, increasing layer coverage does not necessarily improve results, and attention only injections remain the most effective and stable option in this setup.

3.4 LoRA Efficiency Comparison

LoRA finetune significantly improves efficiency by reducing computational load, training time, and storage requirements. The comparisons in training time, GPU memory usage, and model size are summarized below:

	Training time	GPU memory usage (MiB)	Model size(byte)
Gen2Res finetune	44:11	10452	527,068,365
LoRA Att	23:58	5498	200,129
LoRA Att+Res	36:28	7052	2,070,937
LoRA Att+Emb	28:51	7332	984,965

The results demonstrate that LoRA finetuning offers substantial efficiency gains over full parameter training. Gen2Res finetuning took over 44 minutes with peak GPU memory usage of 10.5 GB and produced a model over 500 MB in size. In contrast, LoRA attention only tuning reduced training time and memory usage by nearly half and required only 200 KB of storage. Injecting LoRA into additional layers increased resource usage moderately. Att+Res required the most memory and time among LoRA variants, while Att+Emb offered a good compromise between overhead and performance. These results highlight LoRA's scalability and suitability for low resource or rapid personalization scenarios.

4. Conclusion

This project successfully replicated the Gen2Res framework and extended it with LoRA based finetune to explore more efficient personalization strategies for blind face image restoration. Through experimentation, we found that LoRA finetune applied solely to attention layers offers strong restoration performance while significantly reducing training time, GPU memory usage, and model size. Specifically, attention only LoRA achieved perceptual quality comparable to full parameter finetune, as reflected in LPIPS scores, despite using fewer parameters and a much smaller storage footprint.

Quantitative evaluations using PSNR, SSIM, and LPIPS showed that personalization improves with larger dataset sizes and higher LoRA scaling factors. However, aggressive scaling or insufficient data can lead to underfitting or overfitting, depending on the configuration. Attempts to extend LoRA to additional layers such as the residual block convolutions or timestep embeddings yielded mixed results: while they sometimes improved PSNR or SSIM, they also introduced artifacts or reduced perceptual quality, as confirmed visually and through LPIPS.

Efficiency analysis further highlights the practicality of LoRA. Compared to the Gen2Res baseline, attention-only LoRA reduced model size by over 99% and nearly halved the training time and memory usage, making it especially suitable for real-world applications with limited computational resources.

In summary, attention based LoRA tuning provides an optimal balance between restoration quality and efficiency. It enables fast and lightweight personalization while preserving identity features, suggesting its potential as a scalable solution for face image restoration.

Limitation and Future Work:

While this project demonstrates the effectiveness of LoRA for efficient image restoration, the current implementation still has limitations. Key hyperparameters such as the LoRA rank r, scaling factor$a$, and the relative weighting between multiple LoRA-injected layers were fixed and not fully optimized. As a result, the restoration quality, especially when combining attention, residual, and embedding layers, may not reflect the full potential of the method. Additionally, the current training setup relies heavily on MSE loss, which does not always align with perceptual quality. Future work should explore alternative or composite loss functions, such as perceptual loss or adversarial loss, to better capture visual realism. Further improvements could also come from systematically tuning LoRA parameters, applying adaptive scaling, and evaluating the approach on more diverse datasets and degradation types for better generalization and robustness.

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